

A NEGATIVE PRICE YOU PAY NOTHING FOR



Combo Orders with Futures

Buying a spread, step by step

An engineer's and product-builder's guide to two-legged combos, atomic execution, and how a spread that costs less than nothing still makes money.

Level: Beginner

Worked example: Crude Mar–Jun

~45 minutes

J. Persson, 2026

How to read this deck

We build one idea at a time and keep re-using a single running example so nothing appears from nowhere.

1

One example, all the way

A WTI Crude March–June spread carries every idea — from first order to final cash.

2

Cash, not abstraction

For each step we show the actual dollars that move in and out of your account.

3

Few words, more pictures

Each slide has a purpose-built diagram instead of a wall of bullet points.

4

Beginner assumptions

We assume the platform guarantees both legs fill together. Why that matters comes first.

Why buy and sell at the same time?

Before any mechanics, the obvious question: *why would anyone deliberately buy one thing and sell another at the very same moment? Because sometimes your view isn't 'up' or 'down.'*

Your bet is on the gap between two prices — not on the market's direction.

Buy one contract and sell a related one, and the shared market move cancels out — leaving only the relationship you actually care about.

NO CRYSTAL BALL

You don't have to predict whether the whole market rises or falls.

RISKS CANCEL

If both legs move together, the shared part nets to roughly zero.

ONLY THE GAP

Your profit depends purely on how the two prices move relative to each other.

KEY IDEA A combined buy-and-sell lets you bet on the relationship between two prices while staying neutral on the **market as a whole**.

Does this happen with ordinary shares too?

The idea isn't unique to oil, or even to futures — investors pair trades in stocks all the time. But futures are where it feels natural, because selling is just as easy as buying.

ORDINARY SHARES

Yes, but it's clunky

Investors do buy one stock and short another — ‘pairs trading.’ But to short a share you must first borrow it: find a lender, pay a fee, and risk being forced to return it early.

Selling is harder than buying.

FUTURES

As easy as buying

A futures contract is symmetric. Opening a short is exactly as easy as opening a long — no borrowing, no lender, nothing to locate first.

That symmetry is why combos are a futures thing.

KEY IDEA The concept exists in equities too — but futures make the sell side effortless, which is why combined orders belong to futures.

First — what actually is a futures contract?

We just said a future is ‘symmetric.’ But that only makes sense once you see what a future really is — and it isn't a barrel of oil, or a thing you own at all.

A futures contract is a promise: agree today to trade a fixed amount at a fixed price on a set future date.

No money and no oil change hands now — you are only locking in the terms of a trade that settles later.

JUST A PROMISE

Nothing is owned or delivered today. You are only agreeing terms now for a date down the road.

STANDARDISED

The amount, grade and delivery date are fixed by the exchange, so every contract is interchangeable.

TWO OBLIGATIONS

One side must deliver and be paid; the other must receive and pay. Every contract carries both.

KEY IDEA A future is a contract, not a possession — an agreement to trade later. That is why you can commit to either side of it without owning anything today.

So buying and selling are two halves of one promise

Because a future is only an agreement, taking either side is the same kind of act. You aren't buying or selling a thing — you're choosing which promise to make.

GOING LONG (BUY)

Promise to receive

You agree to take delivery and pay the fixed price on the date. You gain if the market price ends up above it, and lose if it ends up below.

Nothing to own first — just a promise to pay later.

GOING SHORT (SELL)

Promise to deliver

You agree to deliver and collect the fixed price on the date. You gain if the market price ends up below it, and lose if it ends up above.

Equally just a promise — no share to borrow, no oil to hold.

KEY IDEA Buying and selling a future are mirror images — two equal promises. 'Selling' needs no oil and no borrowing because it is simply the other half of the very same contract.

Do you need to own a June future to sell one?

The single most common beginner question. If you 'sell' a June future, surely you must own one first? Here is the surprise.

No. Selling a future opens a short position.

A short is simply an obligation to deliver later — not something you had to own beforehand.

YOU

SELL 1 June

now short -1



COUNTERPARTY

BUY 1 June

now long +1

Neither side owned a June future beforehand — the agreement creates both positions at once.

KEY IDEA Selling a future doesn't require owning one. It opens a short — an obligation you can later close by buying it back.

But then who delivers the oil?

If you sold a June contract and never bought any oil, wouldn't you be forced to hand over 1,000 barrels at expiry? Where would they even come from?

IF YOU HELD TO EXPIRY

Delivery is real

A contract carried all the way to its delivery date does settle — physically (actual barrels) or in cash, depending on the contract.

Very few traders ever go this far.

WHAT TRADERS ACTUALLY DO

Just close early

Before expiry you do the opposite trade — buy back the June you sold. The two cancel, the obligation vanishes, and no barrels change hands.

You source nothing; you just buy the contract back.

KEY IDEA You exit by buying the contract back before delivery — so a spread trader never has to find, store, or deliver a single barrel.

Are you guaranteed to be able to sell?

So if you can sell without owning, are you also guaranteed to sell whenever you want, at the price you want? Two different things hide inside that one question.

WHAT IS GUARANTEED

The price holds

A clearing house stands between buyer and seller and guarantees the contract is honoured. Your counterparty cannot simply walk away from the agreed price.

No default risk on the price you locked.

WHAT IS NOT

Exiting on demand

A resting order to sell at your price only fills if a buyer meets it. A market order fills now — but at whatever price the market offers, not one you choose.

Getting out at your price is never promised.

KEY IDEA The guarantee is that the agreed price will be honoured — not that you can exit whenever, at whatever price. That gap is what the rest of this deck is about.

The journey — six stages

01

The problem

Why trading two contracts separately is dangerous

02

The combo

One instrument, two legs, filled atomically

03

Negative cost

How a spread's price can drop below zero

04

The money

Margin and the daily cash that changes hands

05

The profit

Selling a negative spread back for real dollars

06

What's next

The liquidity gap that implied orders solve

KEY IDEA You are here to reach Stage 05 — earning cash on a spread you bought below zero.

01

SECTION 01



The Problem

Before combos existed, wanting two contracts at once was a gamble. Here is exactly what went wrong — and why it needed fixing.

Futures in 30 seconds

A futures contract is a binding agreement to buy or sell something at a fixed price on a future date. You don't pay the value up front — you commit to the price.

Underlying	WTI Crude Oil
Contract size	1,000 barrels
Quoted in	US\$ per barrel
Delivery month	e.g. March 2026

WHY SIZE MATTERS

\$1 / barrel move



\$1,000 per contract

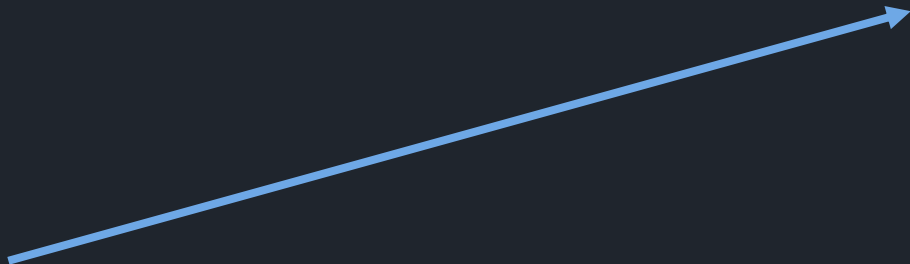
KEY IDEA 1 contract = 1,000 barrels, so every \$1.00 price move is worth \$1,000. Hold this number — we use it constantly.

The trade you actually want isn't a direction

Many traders have no view on whether oil goes up or down. Their view is about the gap between two delivery months — March versus June.

A DIRECTIONAL BET

“Oil will rise.”



Exposed to the whole market. A war, an OPEC cut, a recession — anything moves your P/L.

A RELATIONSHIP BET (our focus)

“March will gain on June.”



*If both rise \$5, you barely care.
You only care that the gap
between them changes.*

KEY IDEA A spread trade bets on the relationship between two contracts — not on the market's direction.

The naive approach: two separate orders

To get “long March, short June,” you could just send two independent orders and hope they both fill. Watch what the clock does to that plan.

BUY**March Crude**target **\$80.00***Order A — sent to the market***SELL****June Crude**target **\$80.50***Order B — sent separately*

THE HOPE

Both fill in the same instant at the prices you wanted, giving a spread of exactly $-\$0.50$.

The reality: the two orders are independent events. Nothing ties them together in time or price.

KEY IDEA Two independent orders can fill at different moments — and the market never waits for you.

Leg risk: a worked example

You send both orders at 10:00:00. Order A fills instantly. Order B lands 400 milliseconds later — and in that sliver of time, June ticks down.



KEY IDEA You paid \$0.40 more than planned — \$400 per spread — purely from the gap between the two fills.

That slippage is real money

\$0.40 sounds tiny. Multiply by the contract size and the number of spreads, and leg risk becomes a serious, recurring tax on every trade.

\$0.40	×	× 1,000	×	× 10	=	\$4,000
worse per barrel		barrels per contract		spreads in the trade		gone before you start

And that assumes the second leg fills at all. Sometimes it doesn't — which is far worse.

KEY IDEA Leg risk scales with size. On a real position it is the difference between a good idea and a losing one.

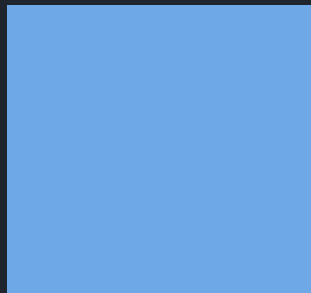
The worst case: a naked leg

Now imagine Order B never fills — no seller shows up at your price. You are left holding only March: a raw, one-sided bet on the whole oil market.

INTENDED: HEDGED



+March



-June

Risks cancel. A market-wide move barely dents you.

ACTUAL: NAKED LONG



+March



(no June)

If oil drops \$3 overnight, that's -\$3,000 per contract of pure, unwanted exposure.

KEY IDEA A half-filled spread isn't a spread — it's an accidental outright position you never meant to hold.

The fix, in one word: atomicity

Every problem so far comes from the two legs being separate. The cure is to make them inseparable — one order that fills completely or not at all.

ALL



Both legs fill together, at one agreed spread. You get exactly the position you asked for.

NOTHING



If the platform can't fill both at once, it fills neither. You are never left half-exposed.

KEY IDEA Atomic execution = both legs or neither. This single guarantee removes leg risk entirely.

02

SECTION 02



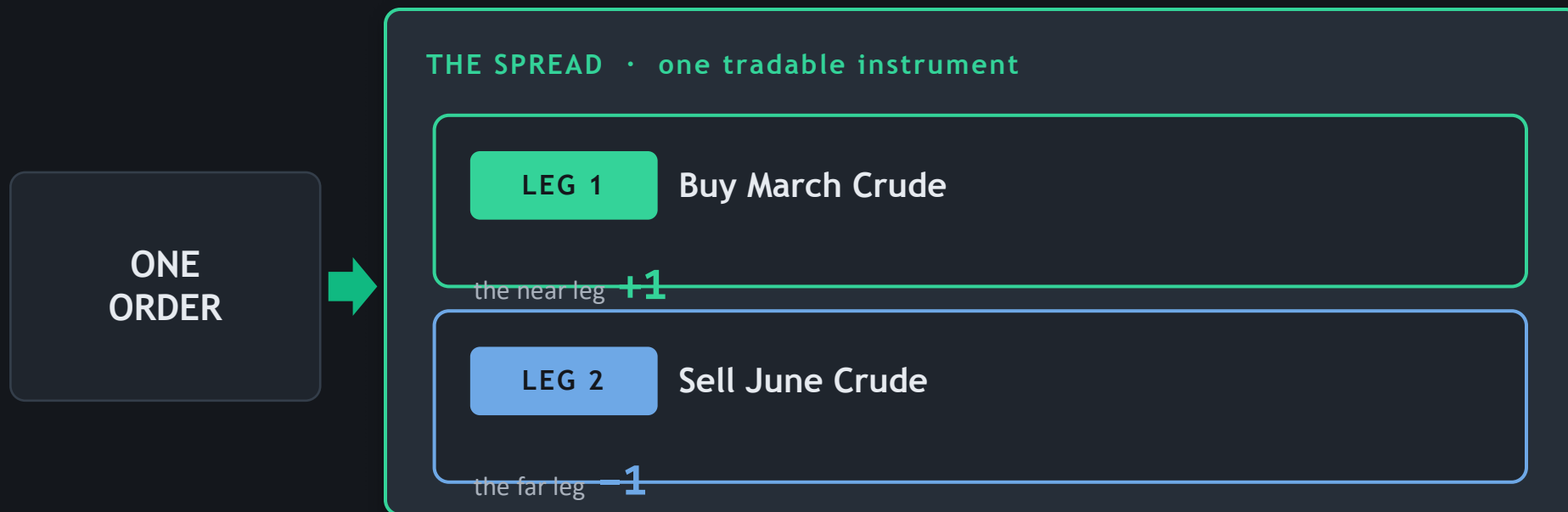
Anatomy of a Two-Legged Combo

One order, two legs, a single price. We start with two legs because it is the simplest place to see the whole machine working.



A combo order is one instrument made of legs

Instead of two order tickets, you place a single order for a purpose-built instrument: the spread. It bundles the legs and the platform quotes it as one price.



KEY IDEA You trade the spread as if it were a single product. The legs come along automatically and atomically.

Our example: the Crude March-June spread

A calendar spread pairs two delivery months of the same commodity. Same oil, same contract size — only the delivery date differs.

NEAR LEG

March Crude

Delivers first — the “front” month

- Same underlying: WTI Crude
- Same size: 1,000 barrels
- Earlier delivery date

FAR LEG

June Crude

Delivers later — the “back” month

- Same underlying: WTI Crude
- Same size: 1,000 barrels
- Three months later

KEY IDEA “Crude Mar–Jun” just means these two months, traded as one spread. That's the whole vocabulary.

Convention: “buy the spread” = buy the near leg

This trips up everyone at first, so memorise it once: buying a calendar spread means buying the near month and selling the far month.

BUY the Mar-Jun spread

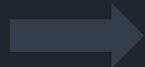


BUY +1 March

and

SELL -1 June

SELL the Mar-Jun spread



SELL -1 March

and

BUY +1 June

KEY IDEA Buy the spread = long the near, short the far. Selling it is the exact mirror.

The spread's price = Near – Far

The spread has its own single price, and it is defined simply as the near-month price minus the far-month price.

$$\text{Spread} = \text{Price}_{\text{March}} - \text{Price}_{\text{June}}$$

When Near > Far

Spread is **positive (+)**

The market is in backwardation.

When Near < Far

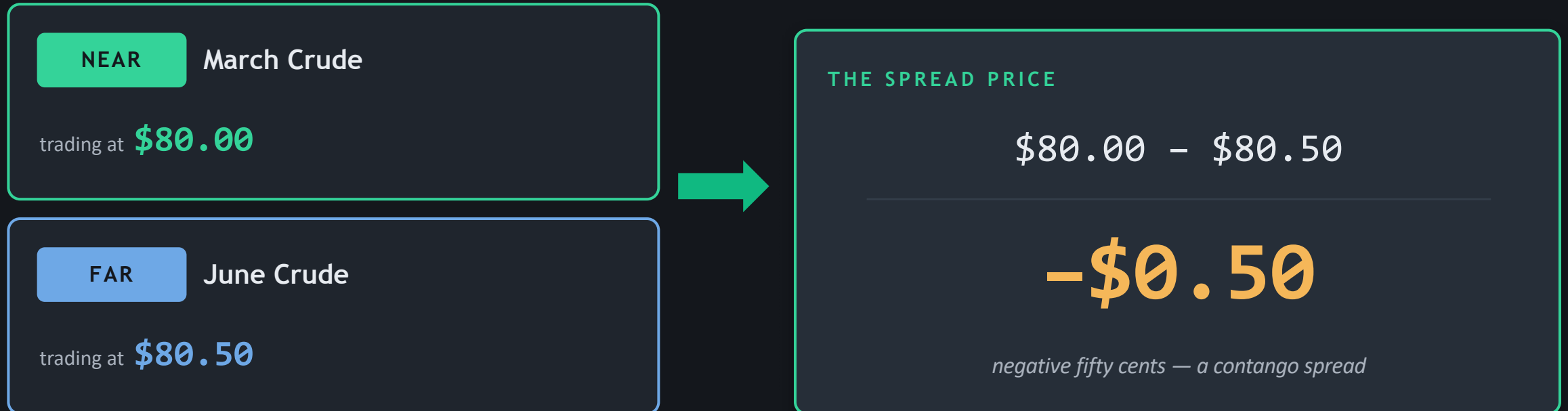
Spread is **negative (-)**

The market is in contango — our case.

KEY IDEA One subtraction defines everything that follows. A negative result is normal, not an error.

Worked example: quoting our spread

Plug in today's prices. March trades at \$80.00, June at \$80.50. The spread price falls straight out of the formula.



KEY IDEA Our spread is quoted at $-\$0.50$. Next we confront what a negative price actually means.

The atomicity guarantee, stated precisely

This is the promise we assume the platform makes. It is what turns two risky orders into one safe instrument.

1

Single price

You agree on one number — the spread — not two leg prices. Here, $-\$0.50$.

2

Both or neither

The fill creates both legs at once, or creates nothing. No partial states exist.

3

Guaranteed ratio

Exactly +1 March against -1 June. The hedge can never come out lopsided.

KEY IDEA From now on we lean on this guarantee completely — it is what makes a spread a single, safe thing to trade.

Placing the order: a walkthrough

Putting it together, here is the single ticket you send and exactly what the platform does with it.

ORDER TICKET

Instrument	Crude Mar–Jun
Action	BUY the spread
Quantity	1 spread
Limit price	-\$0.50
Order type	All-or-none (atomic)



WHAT THE PLATFORM DOES

- 1 Finds a counterparty willing to take the other side of the whole spread at -\$0.50.
- 2 In one atomic action, books +1 March and -1 June to your account.
- 3 No premium changes hands — only a margin deposit is set aside (next section).

KEY IDEA One ticket, one price, one atomic fill. You now hold the spread at -\$0.50.

03

SECTION 03



The Negative-Cost Puzzle

-\$0.50 is a price you pay by receiving. This section makes that sentence feel obvious instead of impossible.

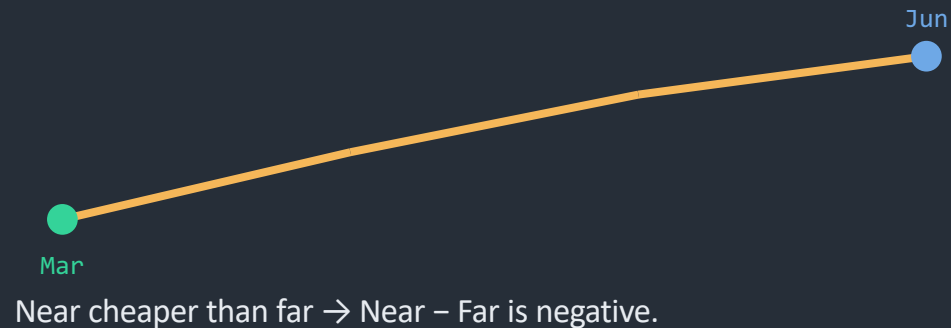


Combo Orders with Futures

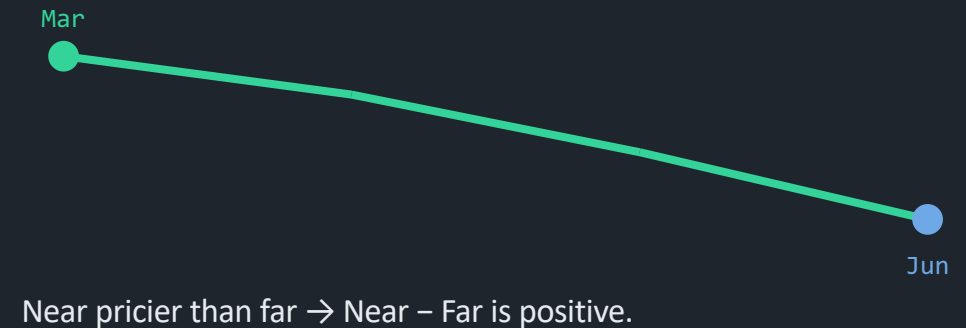
Why a spread can be negative

The sign of the spread just reflects the shape of the forward curve — whether later oil is more or less expensive than nearer oil.

CONTANGO (our case)



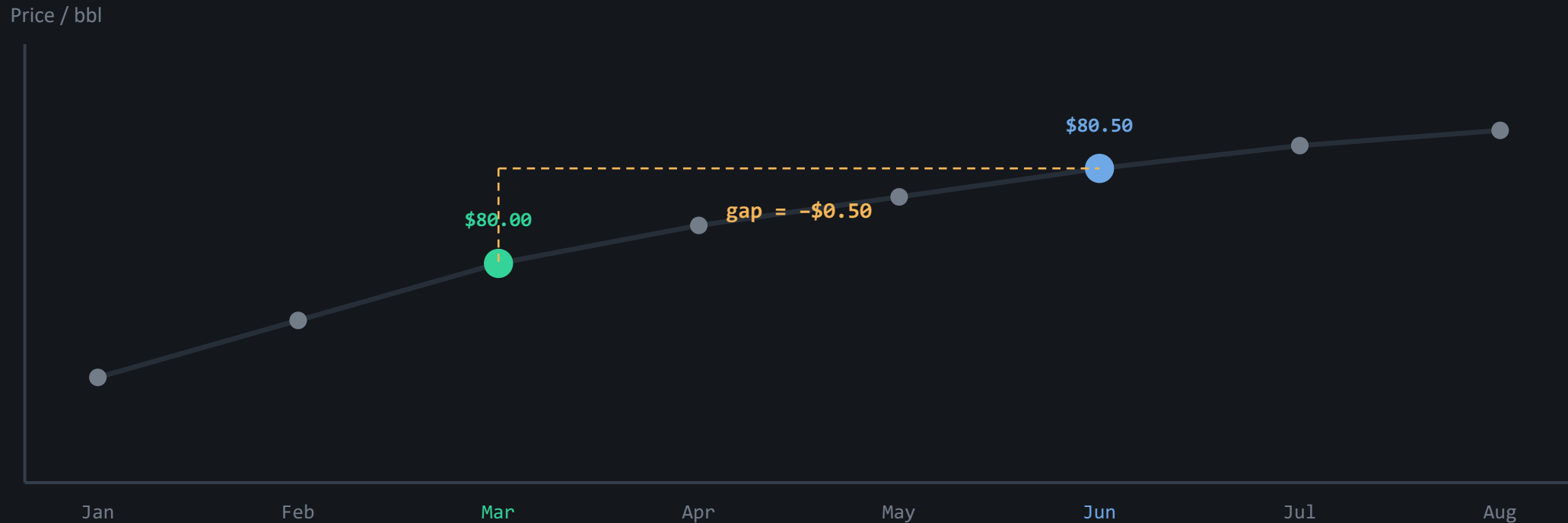
BACKWARDATION



KEY IDEA Contango (rising curve) makes our spread negative. It's a market condition, not a mistake.

Reading the forward curve (illustrative)

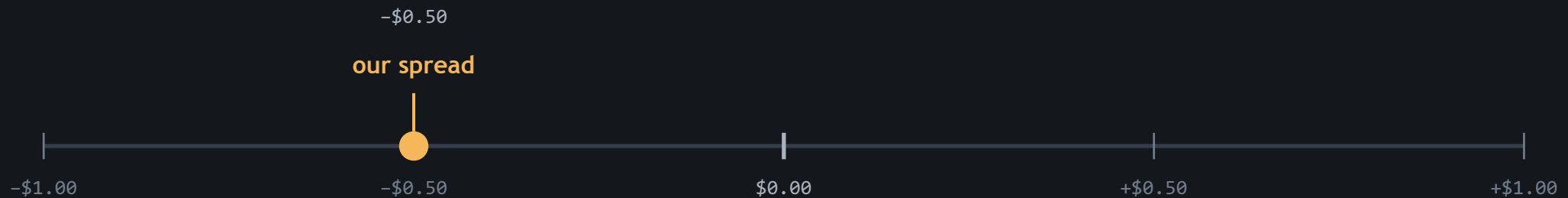
Here is a stylised Crude curve. Each dot is one delivery month's price. Our spread is just the vertical gap between the March and June dots.



KEY IDEA The spread is one slice of the curve. Trading it is a bet on how that slice's slope will change.

Reading a negative spread price

Put the spread on a number line. Zero means the two months are equal; left of zero is contango, right is backwardation.



CONTANGO ← near cheaper

near pricier → BACKWARDATION

The move that pays us: **the marker slides right** — from -\$0.50 toward zero and beyond.

KEY IDEA Our position lives at -\$0.50. We profit if it travels rightward along this line.

“Buying at a negative price?!”

With stocks, buying always costs money. So a negative price feels impossible. The confusion comes from a hidden assumption — let's expose it.

THE STOCK INTUITION

“Buy 1 share at \$80”

You hand over \$80 today and own an asset. Price = cash out of your pocket, always positive.

Hidden assumption: price = money paid up front.

THE FUTURES REALITY

“Buy the spread at −\$0.50”

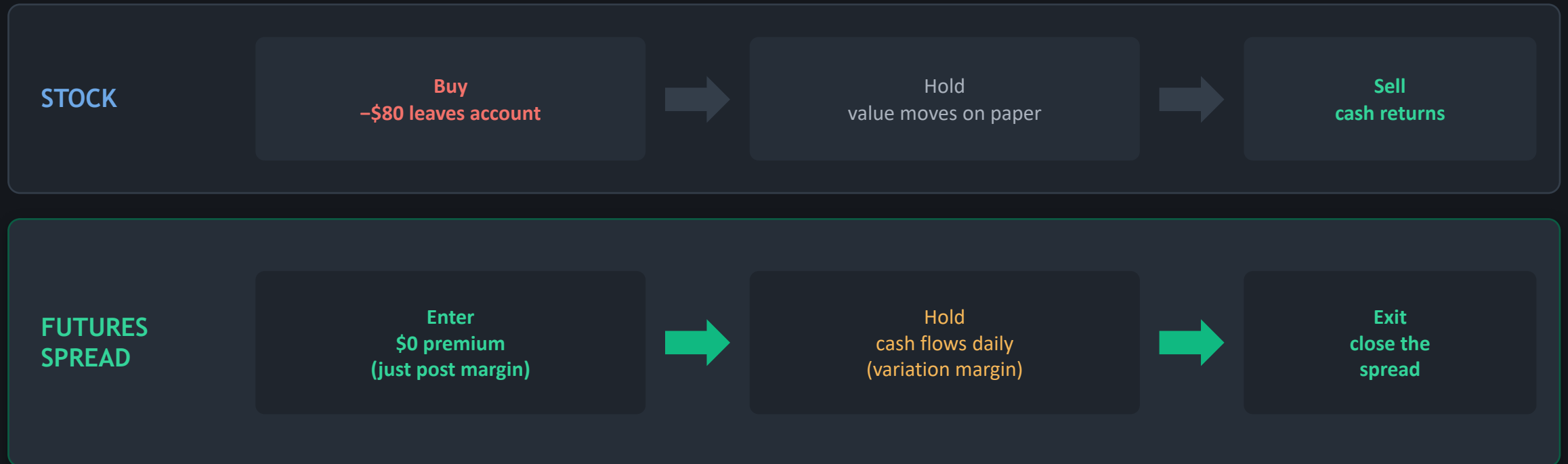
You pay nothing up front. The price is the level you're locking in, not cash you hand over.

Drop the assumption and −\$0.50 stops being paradoxical.

KEY IDEA A futures price isn't cash paid — it's the agreed level. That's why it can sit below zero.

Futures have no purchase price

Compare the two cash timelines. This is the single most important reframing in the whole deck.



KEY IDEA There is no up-front payment to be negative about. -\$0.50 only sets the level you'll be measured from.

At entry, what actually changes hands?

Let's be concrete about day one. You buy the spread at $-\$0.50$. Here is every dollar that moves.

Premium / purchase price

\$0

Unlike a stock or an option, nothing is paid for the position itself.

Initial margin

$\approx \$1,000$

A refundable good-faith deposit, held by the clearing house. Still your money.

Money you 'lose'

\$0

Posting margin is not a cost — you get it back when you close.

KEY IDEA Entering a $-\$0.50$ spread costs you nothing but a deposit you'll get back. Truly.

Worked example: entering the -\$0.50 spread

Your account before and after the fill. Notice the balance barely changes — the margin is ring-fenced, not spent.

BEFORE THE FILL			AFTER THE FILL	
Cash balance	\$25,000		Cash balance	\$25,000
Held as margin	\$0	→	Held as margin	\$1,000
Available	\$25,000		Available	\$24,000
Position	none		Position	+Mar / -Jun

KEY IDEA Cash balance is unchanged. \$1,000 simply moved from ‘available’ to ‘reserved’ — and you now hold the spread.

04

SECTION 04



Follow the Money

If nothing is paid up front, where does profit or loss actually live? In a daily cash flow called variation margin.



Combo Orders with Futures

Two kinds of cash in a futures account

Only two money mechanisms matter for us. Keep them separate in your head and everything else follows.

INITIAL MARGIN

A deposit

- Paid once, when you open the position
- Refundable — returned when you close
- Sized to cover a bad day, not to be spent
- Think: a security deposit on a rental

VARIATION MARGIN

The daily settling

- Cash moves every single day
- Follows the spread's daily change
- Credited to you on good days, debited on bad
- This is where your real profit lands

KEY IDEA Initial margin is a deposit you park. Variation margin is the daily cash that actually is your P/L.

The multiplier: turning \$/bbl into dollars

Spread prices are quoted in dollars per barrel. To get account dollars, multiply the price change by the contract size.

$$\text{spread change (\$/bbl)} \times 1,000 \text{ bbl} = \text{account dollars}$$

\$0.01

1 tick

= \$10

\$0.20

a small move

= \$200

\$0.70

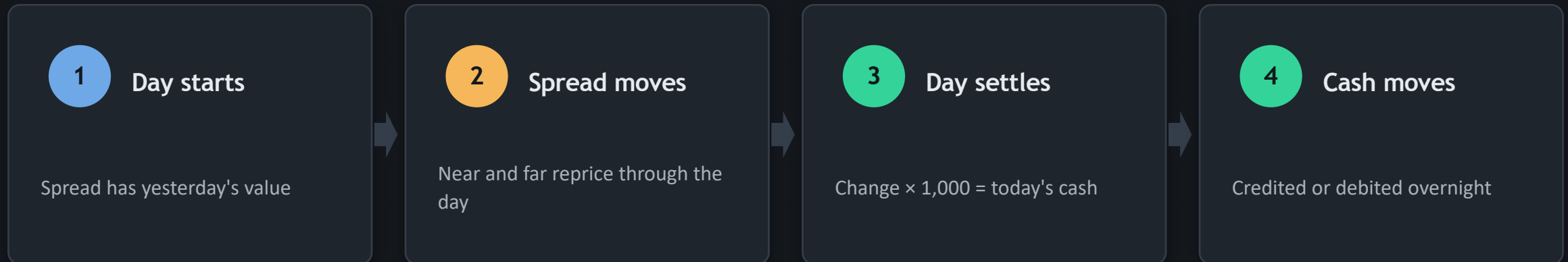
our full trade

= \$700

KEY IDEA Every \$0.01 of spread movement is \$10. Our whole trade will hinge on a \$0.70 move = \$700.

Variation margin: settled every day

Each evening the clearing house re-prices your spread and moves cash so your paper gain or loss becomes real that day. No waiting until you close.

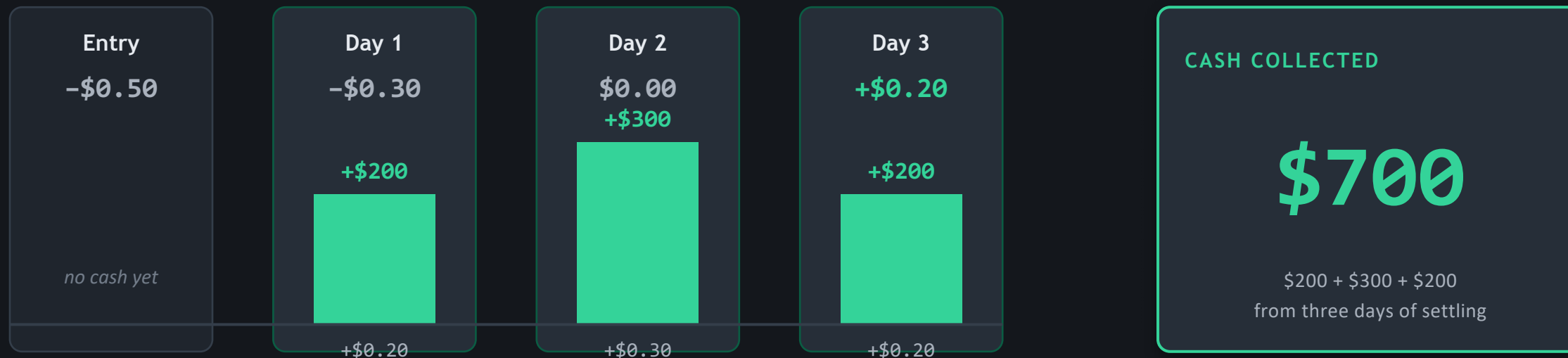


↻ and the loop repeats every trading day until you exit

KEY IDEA Your profit doesn't wait for the exit — it trickles in (or out) as cash, one day at a time.

Day-by-day cash flow of our spread

We enter at $-\$0.50$ and the spread drifts up over three days. Watch the daily cash pile up via variation margin.



KEY IDEA The spread rose $\$0.70$ in total. At $\$1,000$ per point, that's $\$700$ — delivered as daily cash, not a lump sum.

05

SECTION 05



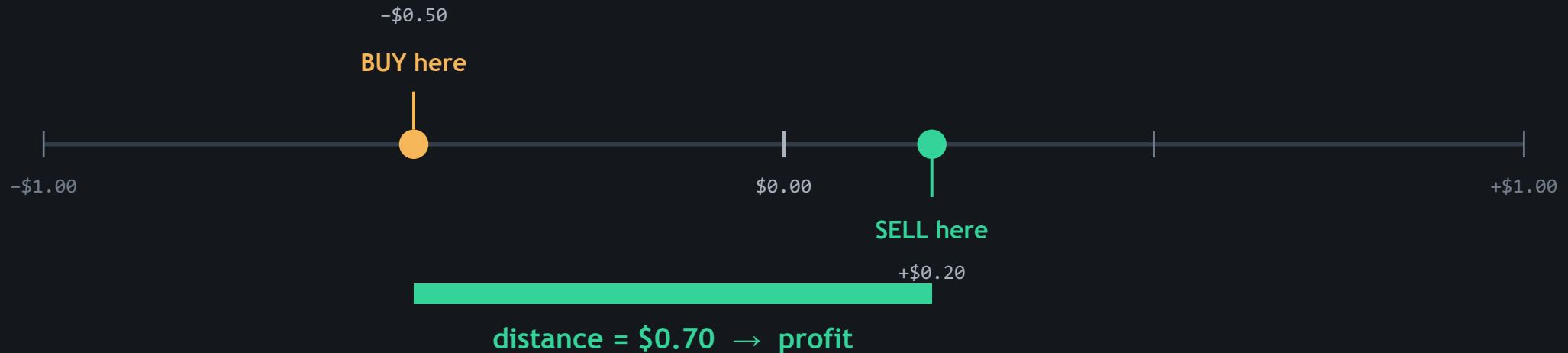
Making Money on a Negative Spread

Now the payoff. We bought at $-\$0.50$, we sell back higher, and we trace every dollar that lands in the account.



The plan: buy low, sell high — even below zero

Profit doesn't care about the sign of the price. It only cares about the distance between where you bought and where you sold.



KEY IDEA $-\$0.50$ up to $+\$0.20$ is a $\$0.70$ gain. 'Buy low, sell high' works perfectly well entirely below zero.

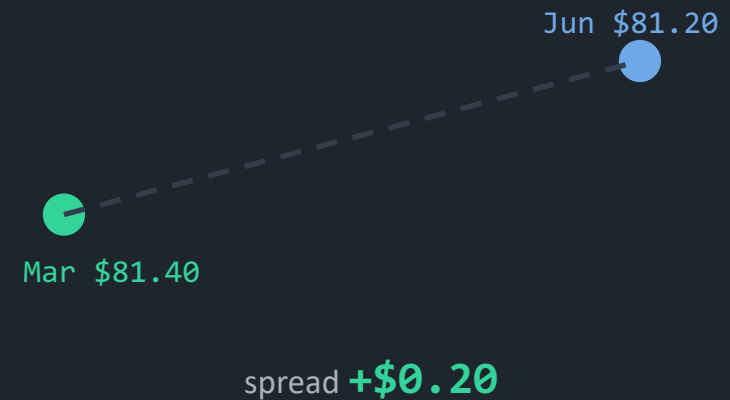
What has to happen: the curve shifts

For the spread to rise, March must gain on June — the curve tilts from contango toward backwardation. Here's the before and after.

BEFORE — at entry



AFTER — at exit



KEY IDEA March overtook June. The gap flipped from -\$0.50 to +\$0.20 — exactly the move we were positioned for.

Closing: sell the spread back

To exit, you do the mirror image of the entry — one atomic order that sells the spread and flattens both legs at once.

YOU HOLD

+1 March -1 June

bought as a spread at $-\$0.50$

SELL 1 spread @ $+\$0.20$

→ sells March, buys back June, atomically

RESULT

flat · no position

Both legs are gone. Your market risk is zero again.

The \$1,000 margin deposit is released back to 'available'.

KEY IDEA Selling the spread closes both legs atomically — the same guarantee that protected you on the way in.

Total P/L in one line

The whole trade collapses to a single calculation: how far the spread travelled, times the contract size.

$$(+\$0.20 - (-\$0.50)) \times 1,000 = +\$700$$

exit price

+\$0.20

minus entry

-(-0.50)

= move

+\$0.70

× 1,000 bbl

+\$700

KEY IDEA Subtracting a negative entry adds it back — that's why the gap is a full \$0.70, and the profit is \$700.

The physical money, from start to finish

Here is the complete cash trail through your account — nothing hidden, every figure adds up.

WHEN	WHAT HAPPENS	CASH NOTE
At entry	Post initial margin (reserved, not spent)	-\$1,000 <i>held</i>
Day 1	Variation margin credited	+\$200 <i>cash in</i>
Day 2	Variation margin credited	+\$300 <i>cash in</i>
Day 3	Variation margin credited	+\$200 <i>cash in</i>
At exit	Initial margin released back	+\$1,000 <i>returned</i>
NET CASH PROFIT (deposit in and out cancels)		+\$700

Being honest: the losing case

The same mechanics run in reverse. If the spread falls instead of rises, variation margin flows out of your account.



$$(-\$0.90 - (-\$0.50)) \times 1,000 = -\$400$$

KEY IDEA Negative-cost doesn't mean risk-free. The spread can widen against you, and that cash really leaves.

The full lifecycle, end to end

Every idea in the deck, on one timeline — from the order ticket to the cash in your account.

Place combo

Buy Mar–Jun at $-\$0.50$,
atomically

1

Hold and settle

Daily variation margin, $+\$700$
total

3

Cash out

Margin released, $+\$700$ profit
kept

5

Post margin

$\$1,000$ reserved, $\$0$ premium

2

Sell to close

Exit at $+\$0.20$, both legs flat

4

KEY IDEA One atomic buy, a few days of daily cash, one atomic sell — and a negative-cost spread became $+\$700$.

06

SECTION 06



Where This Goes Next

You now understand a two-legged combo completely. One practical problem remains — and it points to the next presentation.

What you can now explain to someone else

If these five sentences feel obvious, the deck did its job.

✓ Why trading two legs separately is dangerous — leg risk and naked exposure.

✓ How atomic execution fixes it: both legs at one spread price, or neither.

✓ Why a spread's price can be negative, and why that is not a paradox.

✓ That entering costs \$0 up front — only a refundable margin deposit.

✓ How daily variation margin turns a \$0.70 move into \$700 of real cash.

The one problem we left open

Our whole story assumed a counterparty was waiting to take the exact spread we wanted. Often, nobody is quoting the spread directly — only the individual months trade.

WHAT WE ASSUMED

Someone is quoting Mar–Jun at $-\$0.50$, ready to trade the spread as one unit.

Mar–Jun spread book: liquid

WHAT OFTEN HAPPENS

The spread book is empty — but March and June each trade actively on their own.

Spread book:
empty

Outright months:
liquid

KEY IDEA The atomic combo is only useful if someone will trade the spread. Sometimes the liquidity is in the legs instead.

The next chapter: implied orders

Implied orders are how an exchange builds a spread trade out of the separate month markets — and vice-versa — so your atomic combo can fill even with an empty spread book.

March
market

+

June
market



an IMPLIED Mar–Jun spread

The exchange synthesises a spread quote from the two outright books — no direct spread trader required.

That's a whole topic of its own. We'll cover how implied pricing and matching work in a dedicated follow-up presentation.

KEY IDEA Implied orders bridge the legs and the spread. That's where the next deck begins.

YOU CAN NOW TRADE A SPREAD — AND EXPLAIN WHY IT WORKS

Bought at less than nothing.
Sold for \$700.

Two legs, one atomic order, and a clear-eyed view of every dollar that moved.

■ Next up: Implied Orders — filling a spread from the legs.